

Relationships Between Child-Reported Activity Level and Task Orientation and Parental Attention-Deficit/Hyperactivity Disorder Symptom Ratings

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ABSTRACT: *Objective:* This study examines whether elementary school-aged children can report behaviors relevant to assessing symptoms of attention-deficit/hyperactivity disorder (ADHD). *Methods:* Interviews were conducted with 120 children aged 6 to 12 years and their parents across 3 waves as part of a longitudinal cohort study of ADHD detection and service use. Child self-reports obtained through the Dimensions of Temperament Scale-Revised-Child (DOTS-R-C) were correlated with parent-reported ADHD symptoms, which were assessed through DSM-IV-based instrument ratings obtained concurrently and 5 years later. *Results:* The Dimensions of Temperament Scale-Revised-Child subscales Activity Level and Task Orientation demonstrate adequate internal consistency after eliminating items requiring reverse scoring. Children's self-reports of Task Orientation Problems correlate with their parents' concurrent reports of inattention, $r(117) = .23, p < .05$, and with parents' Wave 3 reports of inattention, $r(118) = .25, p < .01$ as well as hyperactivity, $r(118) = .25, p < .01$. Children's self-reports of Activity Level correlate with their parents' concurrent reports of hyperactivity, $r(117) = .21, p < .05$, as well as Wave 3 reports of hyperactivity/impulsivity, $r(118) = .37, p < .001$ and inattention, $r(118) = .23, p < .05$. *Conclusions:* Findings suggest that children may be capable of producing meaningful self-reports of Activity Level and Task Orientation. We propose that the development of child-friendly self-report instruments targeting ADHD symptoms is merited to facilitate the collection of child input during ADHD assessments.

(*J Dev Behav Pediatr* 31:233–237, 2010) **Index terms:** ADHD, child self-report, temperament, hyperactivity, inattention.

Current practices in the assessment and evaluation of children with attention-deficit/hyperactivity disorder (ADHD) rely largely on adult reports of children's behavior. The American Academy of Child and Adolescent Psychiatry's Practice Parameters for the Assessment and Treatment of Children and Adolescents with ADHD¹ call for the use of interviews and rating scales to be completed by the patient's parent and teachers. A child interview is recommended to determine the presence of symptoms suggestive of additional comorbid disorders but not to refute or confirm the diagnosis of ADHD, because young children are considered to be unaware of their own symptoms, and adolescents are expected to minimize their symptoms' significance.²

Clinical reluctance to include child self-reports in the diagnostic assessment process for ADHD may reflect con-

cerns about limited empirical support for available assessment tools. A recent review of methods for assessing externalizing disorders cautions that the validity and reliability of widely used child self-report measures is not well established, although some studies suggest that the Youth Self-Report and the Children's Action Tendency Scale are adequate measures.³ In contrast, in the context of temperament research, children have been found to be valid reporters of their own externalizing behavior. For example, the Revised Dimensions of Temperament Scale-Child (DOTS-R-C) has satisfactory construct and concurrent validity,^{4,5} and high rates of agreement between parental and child ratings on the DOTS-R-C have been reported.⁶

Including child self-reports in the ADHD assessment process may be beneficial for several reasons. First, it may help with the process of treatment engagement because increased direct communication between physician and child has been shown to improve parental satisfaction with care and adherence to treatment regimens.^{7,8} Furthermore, child self-ratings may tap into aspects of psychopathology not elicited by parental assessment tools based on the Diagnostic and Statistical Manual, 4th edition (DSM-IV). In a study examining the relationship between childhood temperament and adolescent ADHD symptom trajectories, childhood self-ratings of difficult temperament on the DOTS-R-C correlated with parent-rated externalizing symp-

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toms in adolescence and were associated with less improvement of hyperactivity/impulsivity symptoms in adolescence.⁹ Subsequent analyses showed that the prediction of ADHD symptom persistence was only related to 2 of the 6 components comprising the difficult temperament variable, Activity Level, and Task Orientation. Here, we follow-up on these findings and examine whether elementary schoolchildren can report behaviors relevant to the assessment of ADHD symptoms. We hypothesize that children's self-reports of Activity Levels and Task Orientation on the DOTS-R-C can be interpreted as self-reports of certain ADHD symptoms. Specifically, we determine the internal consistency of children's self-reports of Activity Levels and Task Orientation on the DOTS-R-C and examine their concurrent and predictive validity through correlations between child self-reports and parent ratings of ADHD symptoms concurrently obtained and 5 years later.

METHODS
Procedure

This research uses data from a longitudinal study on attention-deficit/hyperactivity disorder (ADHD) detection¹⁰ and has been approved by the University Institutional Review Board and the school district research director. Informed consent was obtained from parents and assent from children before study participation.

As shown in Figure 1, data were collected in 3 waves between the years 1999 and 2006 as follows. At Wave 1, parent ratings of the Swanson, Nolan, and Pelham, Version IV¹¹ and treatment information were obtained to identify students at high risk for ADHD. At Wave 2, parents com-

pleted diagnostic interviews assessing ADHD and disruptive disorders, using the Diagnostic Interview Schedule, Version 4,¹² and children completed the Dimensions of Temperament Scale-Revised-Child (DOTS-R-C).^{12,13} At Wave 3, the Vanderbilt ADHD Diagnostic Parent Rating Scale was used to assess symptoms of ADHD.¹⁴

Participants

By using school district registration records, we identified 12,009 elementary school students enrolled in kindergarten through fifth grade in a North Central Florida public school district during the academic year 1998–1999. To achieve adequate power for future data analyses, 3158 of these children were selected using a gender-stratified random design, over-sampling girls by a margin of 2 to 1 to ensure adequate representation. Only 1 child per household was eligible for selection to ensure subject independence. Children were eligible for the study if they lived in a household with a telephone, were not receiving special education services for mental retardation or autism, and were from white or African-American backgrounds. Children from other ethnic backgrounds were excluded because they comprised <5% of the total student population.

A total of 1615 families completed telephone screening for ADHD risk at Wave 1 (79% participation rate). By using parental reports and standardized parent and teacher behavioral ratings, youth were identified as being at high risk for ADHD if a parent reported that their child had a professional ADHD diagnosis or was receiving formal ADHD treatment, had concerns reported by family or school professionals that the child may have

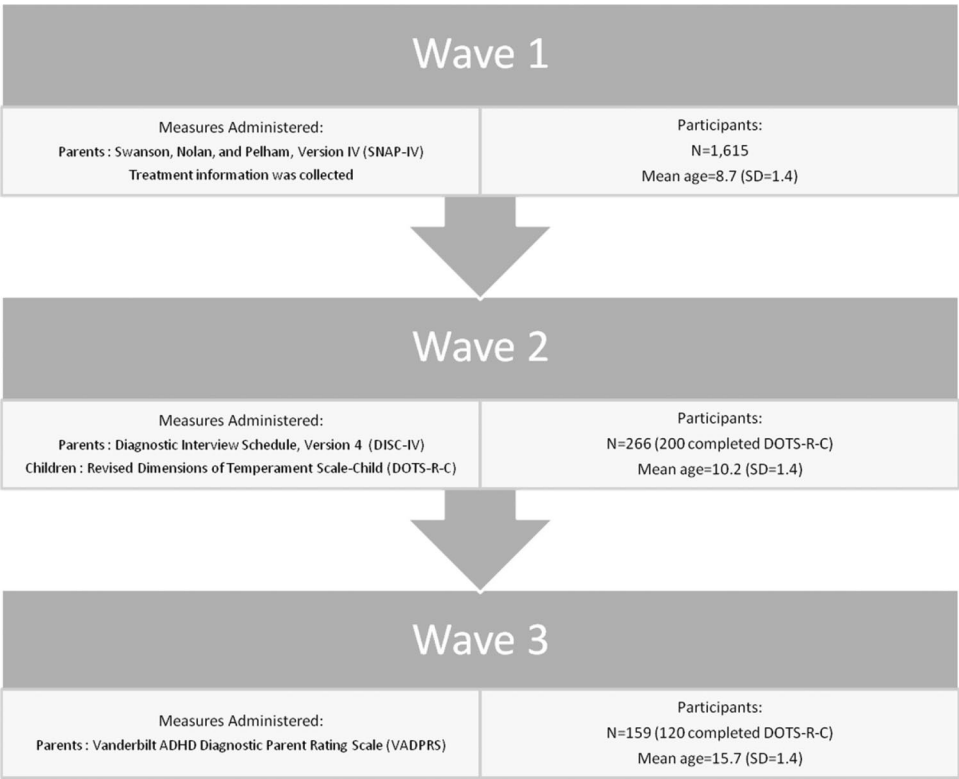


Figure 1. Outline of study participants and measures administered at each wave.

ADHD, or had concerns reported about a behavioral problem if any parent ADHD subscale scores were elevated 2 SDs above age and gender norms. In Wave 2, those children considered at high risk for ADHD ($N = 266$) and their parents underwent diagnostic and service use assessments at participants' homes or community locations; of those children, 200 were aged 8 years or older and eligible to complete the DOTS-R-C. Wave 3 consisted follow-up interview 5 years later to collect data on service use and behavior, which was completed by 159 parents, and Wave 2 DOTS-R-C data were available from 120 adolescents. A detailed description of the study population and the sampling method is described elsewhere.¹⁰

This analysis includes 120 youths with complete data sets from all 3 study waves who had completed the Revised DOTS-R-C as elementary school students. This sample represents 60% of those who were eligible to complete the DOTS-R-C at Wave 2, and there were no differences in Wave 2 temperament scores between those children who participated in Wave 3 and those who did not. This sample consists of 66 girls (55%), 43 African Americans (36%), and 63 children (53%) qualifying for free or reduced school lunch, an indicator of poverty. Furthermore, 71 (59%) met DSM-IV criteria for ADHD, 47 (39.5%) for oppositional defiant disorder, and 14 (12%) for conduct disorder. Average age was 8.7 (SD, 1.4) at Wave 1, 10.2 (SD, 1.4) at Wave 2, and 15.7 (SD, 1.4) at Wave 3.

Measures

Child Sociodemographic Characteristics

Sociodemographic characteristics including the youth's age, gender, lunch status, and race were obtained from school district records and confirmed during interviews.

Child Self-Reports of Activity Level and Task Orientation

The DOTS-R-C was designed to measure the 9 dimensions of temperament originally proposed by Thomas and Chess¹⁵ in children aged 8 years and older.⁴ Windle et al⁴ reported internal consistency coefficients (Cronbach's alphas) for the 9 dimensions that ranged from 0.54 (Rhythmicity-Daily Habits) to 0.81 (Activity Level-Sleep) in a sample of elementary students. The test-retest reliability was 0.64 for hyperactivity scale and 0.55 for the inattention scale. The construct validity of this scale has been supported by factor analytic results confirming the 9 factors. Concurrent validity with measures of perceived cognitive competence, perceived social competence, and general self-worth has been established.⁴ The DOTS-R-C contains 2 subscales that measure constructs relevant to ADHD symptom assessment, Activity Level composed of 7 items (e.g., "I can't sit still for long") and Task Orientation, composed of 5 distractibility items (e.g., "Things going on around me cannot take me away from what I am doing") and 3 persistence items (e.g., "Once I take something up, I stay with it") (see Appendix). A 4-choice response option is given for each item: (1) usually false, (2) more false than true, (3)

more true than false, and (4) usually true. The measure was administered by trained research assistants who read the measure to the children and recorded their responses.

Adult Reports of Child ADHD Symptoms

ADHD symptoms were assessed using standardized parent questionnaires consisting of operationalized DSM-IV criteria for ADHD: the Swanson, Nolan, and Pelham Version IV¹¹ and the Vanderbilt ADHD Diagnostic Parent Rating Scale.¹⁴ To assign research diagnoses of ADHD, parents were administered the Diagnostic Interview Schedule Version IV, which, consistent with DSM-IV diagnostic criteria, inquires about ADHD symptoms and related impairment across school and home settings.¹² Furthermore, we used Diagnostic Interview Schedule Version IV inattentive and hyperactive/impulsive symptom counts, which could each range from 0 to 9, for the correlational analyses.

Data Analysis

The internal consistency of the DOTS-R-C and its Activity Level and Task Orientation Subscales was assessed using Cronbach's alphas. The relationships between children's self-report of activity and Task Orientation on the DOTS-R-C and adult ADHD symptom ratings were assessed through Spearman correlation coefficients between these subscales and parents' concurrent inattention and hyperactivity/impulsivity Diagnostic Interview Schedule Version IV symptom scores, as well as with parents' Wave 3 Vanderbilt ADHD Diagnostic Parent Rating Scale scores 5 years later.

RESULTS

Internal Consistency of the Dimensions of Temperament Scale-Revised-Child

The Dimensions of Temperament Scale-Revised-Child (DOTS-R-C) showed high internal consistency (Cronbach's alphas = 0.80). The original Activity Level and Task Orientation Subscales demonstrated low levels of internal consistency with Cronbach's alphas of 0.59 and 0.57, respectively; however, after eliminating 1 item in each scale that required reverse scoring and produced the lowest item-total correlations (-13 and -35), the Cronbach's alphas for the revised 6-item Activity Level Subscale increased to 0.75 and that of revised 7-item Task Orientation Subscale to 0.66.

Correlations With Concurrent Parental Attention-Deficit/Hyperactivity Disorder Symptom Reports

Children's reports of their Activity Level on the DOTS-R-C significantly correlated with their parents' concurrent reports of hyperactivity/impulsivity on the Diagnostic Interview Schedule Version 4 $r(117) = .21, p < .05$. Children's self-reports of Task Orientation Problems significantly correlated with their parents' concurrent reports of inattention on the Diagnostic Interview Schedule Version 4, $r(117) = .23, p < .05$.

Correlations With Subsequent (Wave 3) Parental Attention-Deficit/Hyperactivity Disorder Symptom Reports

Children's self-reports of Activity Level on the DOTS-R-C significantly correlated with their parents' Wave 3 reports of hyperactivity on the Vanderbilt ADHD Diagnostic Parent Rating Scale, $r(118) = .37, p < .001$, as well as parents' reports of inattention, $r(118) = .23, p < .05$.

Children's self-report of Task Orientation Problems on the DOTS-R-C significantly correlated with their parents' Wave 3 reports of inattention, $r(118) = .25, p < .01$ on the Vanderbilt ADHD Diagnostic Parent Rating Scale as well as parents' Wave 3 reports of hyperactivity $r(118) = .25, p < .01$.

DISCUSSION

Our study indicates that children may be capable of producing meaningful self-reports of Activity Level and Task Orientation. We found small correlations between children's self-reports of Activity Level and Task Orientation and their parents' concurrent reports of hyperactivity/impulsivity and inattention symptoms. These relationships did not attenuate, and children's self-reports predicted parental attention-deficit/hyperactivity disorder (ADHD) symptom reports 5 years later. The observed relationships between child and parent reports were weak but may become more pronounced if child's input is collected through a self-report measure explicitly addressing the full range of the 18 ADHD symptoms rather than just the items included in the DOTS-R-C, which focus on Activity Level and Task Orientation.

Although our findings cannot determine whether the DOTS-R-C facilitates the child's self-report of temperament traits or of ADHD symptoms, they suggest that the DOTS-R-C may serve as a useful example when designing measures that will incorporate children's perspectives into the ADHD assessment and treatment process.¹⁶ This may be important because several instruments enable child self-reports of anxiety or depression^{17,18}; however, no established self-report assessment instruments for childhood ADHD symptoms are currently available, although 1 measure is available for adolescents aged 12 to 18.¹⁹ In addition, pilot work to develop a self-evaluation scale for children with ADHD using a visual format has been reported.² Our data suggest that, if such instruments become available, children may be able to make valuable contributions to the ADHD assessment process. As noted by Wissow and Bar-Din Kimel,⁸ incorporating children's input into diagnosis and intervention planning can help to strengthen the relationship between the patient, family, and treatment provider and may contribute to increasing treatment adherence in²⁰ areas in need of improvement for ADHD care. Further research is merited to develop valid measures for this purpose, using child-friendly language and focusing on pertinent aspects of ADHD that a child can be expected to comprehend.

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Activity Level

I can't sit still for long.

I move around a lot.

If I have to stay in one place for a long time, I get very restless.

Even when I am supposed to be still, I get very fidgety after a few minutes.

I never seem to stop moving.

I never seem to be in the same place for long.

*Eliminated: I often stay still for long periods of time.

Task Orientation

Once I am involved in a task, nothing can distract me from it.

I persist at a task until it's finished.

I stay with an activity for a long time.

If I am doing one thing, something else occurring won't get me to stop.

Things going on around me cannot take me away from what I am doing.

Once I take something up, I stay with it.

I am hard to distract.

*Eliminated: I can always be distracted by something else, no matter what I may be doing.

Book Review

Positive Parenting: Raising Healthy Children from Birth to Three Years

by Alvin N. Eden, MD, New York, NY, Hatherleigh Press, 2007, 288 pp, Paperback, \$15.95.

Nearly 30 years have passed since the initial publication of *Positive Parenting*, and in that time, new guidance has been issued about sudden infant death syndrome, new research has evolved about the importance of early child development, and new health epidemics such as childhood obesity and type II diabetes have emerged. It is no wonder why Dr. Eden, a practicing pediatrician for more than 40 years, felt compelled to update his insightful and humorous work. In doing so, he has yet again created an easy-to-read parenting guide that skillfully walks parents through their child's first 3 years of life.

Immediately on opening his book, parents will undoubtedly be pleased that Dr. Eden's work has the endorsement of Dr. James Stockman III, President of the American Board of Pediatrics, who completed the book's Forward. The book is skillfully broken down into easily manageable chapters. Dr. Eden devotes 4 chapters to the first 12-month-old group, 2 chapters to the 12- to 24-month-old group, and 1 chapter to the 2- to 3-year-old group. Chapter lengths averaging around 30 double-spaced pages should be manageable to even the busiest parents. In addition, the health literacy level makes this book accessible.

Another strength of Dr. Eden's work is the quick reference tables for each age group highlighting typical milestones across all developmental domains. He also provides parents with age-appropriate and easy-at-home activities to promote their child's development. "Your baby's enthusiastic search for knowledge

must be encouraged in every way possible" is thankfully a theme that is repeated throughout the book. Nearly, all activities can be completed with routine household items and none require the parents to purchase the latest and greatest "promote your child's development" toy. Suggestions go a long way toward demystifying ways parents can promote child development in their own homes.

As a pediatrician, I also appreciate Dr. Eden's dedicated space within each chapter to address many of the wide-ranging myths that parents want answered. These include concerns about breast feeding, formula, child care, fevers, and developmental topics. However, I have some concerns about some of his recommendations. Although I certainly appreciate Dr. Eden's experience, some of the newer AAP recommendations contradict some of his suggestions. Minor variations are expected between each practitioner when it comes to advancing infant diets, such as the timing of fruits versus vegetables, and the introduction of high-allergen foods such as eggs. However, I was particularly troubled by his sanctioning of spanking in certain "special occasions" such as "when he's tried to climb out the window" and "riding his tricycle out into the street." Sanctioning even one instance that permits a parent to use corporal punishment sets up a slippery slope that I am reluctant to endorse. Although I understand his intent, I think it is more important that pediatricians communicate to parents a "zero tolerance" approach to physical punishment. He does,

however, offer several other effective discipline strategies and emphasizes that discipline should focus on the "learning experiences of your child, and not simply punishment."

Dr. Eden also spends a portion of his book discussing ways to promote positive self-body image, reduce obesity, and address the negative impact of television on children. His alarming statistics include that two thirds of preschoolers have televisions in their bedrooms and that children watch 18,000 murders on television by the age of 18 years and will make any pediatrician (and hopefully parent) cringe!

Overall, Dr. Eden's updates to *Positive Parenting* are timely, effective, and well done. His global approach to child development and emphasis on engaging parents in the daily lives of their children are the greatest assets to this work. However, before recommending this to parents, I would suggest each provider consider their stance on some of the issues (such as discipline and feeding) described above and address those with parents first. However, in a busy primary care practice, Dr. Eden's work will nicely supplement a pediatrician's efforts to promote child development, address common parental concerns, and actively involve parents in their child's learning.

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